

DR. MEHBUB ALAM

LIST OF PUBLICATIONS

- J1. Mehbub Alam, Mariam Alhashmi, Mohammad Atrouz, Abdulrahman Khader, Ravi Ashok Pashchapur and Abdulhadi Shoufan, "Authentication Protocols for UAV Remote Identification Using Public-Key Infrastructure", *Computers & Security*, Pages: (yet to assign) 2026.
Impact Factor: 6.8 - Q1 (Top 1%)
- J2. Muhammad Owais, Muhammad Zubair, Daniya Najiha Abdul Kareem, Sana Akhtar Naseer, **Mehbub Alam**, Mubashir Ahmad, and Irfan Hussain, "A Comprehensive Review Tracing the Evolution of Volumetric Medical Imaging Analysis from Classic CNNs to Emerging AI-Agents", *Archives of Computational Methods in Engineering*, 2026. **Impact Factor: 12.9 - Q1**
- J3. Sk Mahmudul Hassan, Kumar Sekhar Roy, **Mehbub Alam**, Sudip Chakraborty, Marcello Di Bonito, Ruhul A. Hazarika, "Fusion-Based Machine Learning Approach for Classification of Seabed Defects", *Earth Systems and Environment*, 2026. **Impact Factor: 5.0 - Q1**
- J4. Sk Mahmudul Hassan, Kumar Sekhar Roy, Ruhul A Hazarika, **Mehbub Alam** and Mithun Mukherjee, "Multi-kernel Inception-Enhanced Vision Transformer for Plant Leaf Disease Recognition", in *Scientific Reports*, 15, 30997, 2025. **Impact Factor: 3.9 - Q1**
- J5. Baharul Islam, Nasim Ahmad, **Mehbub Alam**, Sk Mahmudul Hassan, Sudip Chakraborty, Khaled A. Mahmoud, Saad Aljlil and Ruhul A Hazarika, "AquaFusionNet: A Deep Feature Fusion and Attention-Based Approach for Accurate Underwater Defect Detection", in *Earth Systems and Environment*, Pages 1-17, 2025. **Impact Factor: 5.0 - Q1**
- J6. **Mehbub Alam**, Nurzaman Ahmed, Shyamal Ghosh, Rakesh Matam, and Ferdous Ahmed Barbhuiya, "OptiFog: A Framework for Acquiring State Information and Predicting Resource Availability for Task Offloading in Cooperative Fog-Networks", in *IEEE Transactions on Services Computing*, Early Access, Pages 1-13, 2024. **Impact Factor: 5.8 - Q1**
- J7. **Mehbub Alam**, Nurzaman Ahmed, Rakesh Matam, and Ferdous Ahmed Barbhuiya, "Analyzing the Suitability of IEEE 802.11ah for Next Generation Internet of Things: A Comparative Study", in *Ad Hoc Networks, Elsevier*, Vol. 156, Pages 103437, 2024. **Impact Factor: 5.4 - Q1**
- J8. **Mehbub Alam**, Nurzaman Ahmed, Rakesh Matam, Mithun Mukherjee, and Ferdous Ahmed Barbhuiya, "SDN-based Re-configurable Edge Network Architecture for Industrial Internet of Things", in *IEEE Internet of Things Journal*, vol. 10, pages: 16494-16503, 2023.
Impact Factor: 8.2 - Q1
- J9. **Mehbub Alam**, Nurzaman Ahmed, Rakesh Matam, and Ferdous A Barbhuiya, "IEEE 802.11ah-Enabled Internet of Drone Architecture", in *IEEE Internet of Things Magazine*, Vol. 5, Pages 174-178, 2022. **Impact Factor: 5.8 - Q1**
- C1. **Mehbub Alam**, Nurzaman Ahmed, Rakesh Matam, and Ferdous Ahmed Barbhuiya, "ioFog: Prediction-based Fog Computing Architecture for Offline IoT", in *International Wireless Communications and Mobile Computing (IEEE IWCMC)*, Pages 1387-1392, 2021.
- C2. **Mehbub Alam**, Nurzaman Ahmed, Rakesh Matam, and Ferdous Ahmed Barbhuiya, "L3Fog: Fog Node Selection and Task Offloading Framework for Mobile IoT", in *IEEE INFOCOM, IEEE Conference on Computer Communications Workshops (IEEE INFOCOM WORKSHOPS)*, Pages 1-6, 2022.
- C3. Sangeeta Kakati, **Mehbub Alam**, Rakesh Matam, Ferdous Ahmed Barbhuiya, and Mithun Mukherjee, "Mobility-aware Task Offloading in Fog-Assisted Networks", in *IEEE Global Communications Conference (IEEE GLOBECOM)*, Pages 2897-2902, 2022.

- C4. **Mehbub Alam**, Rakesh Matam, and Ferdous Ahmed Barbhuiya, “*OptFog: Optimized Mobility-Aware Task Offloading and Migration Model for Fog Networks*”, in *IEEE International Conference on Advanced Networks and Telecommunications Systems (IEEE ANTS)*, Pages 539-544, 2023.
- C5. **Mehbub Alam**, Nurzaman Ahmed, Rakesh Matam, and Ferdous Ahmed Barbhuiya, “*RedgeX: Meta-learning based Optimal Analytical Model for Programmable Edge Intelligence*”, in *IEEE Wireless Communications and Networking Conference (IEEE WCNC)*, Pages 1-6, 2024.
- C6. **Mehbub Alam**, Rakesh Matam and Ferdous Ahmed Barbhuiya, “*Edge-Mi: Edge-based Microservices for Mobility-aware Task Migration Scheme*”, in *IEEE Future Networks World Forum (IEEE FNWF)*, 2024.
- C7. Nasim Ahmed, Baharul Islam, **Mehbub Alam**, and Sudipta Majumder, “Leveraging LLMs and Deep Learning for Driver Drowsiness Image Categorization”, *IEEE Guwahati Subsection Conference (IEEE GCON)*, 1-6, 2025